Module java.base
Package java.util.stream

Interface IntStream

All Superinterfaces:

AutoCloseable, BaseStream<Integer,IntStream>

```
public interface IntStream
extends BaseStream<Integer,IntStream>
```

A sequence of primitive int-valued elements supporting sequential and parallel aggregate operations. This is the `int` primitive specialization of `Stream`.

The following example illustrates an aggregate operation using `Stream` and `IntStream`, computing the sum of the weights of the red widgets:

```
int sum = widgets.stream()
                 .filter(w -> w.getColor() == RED)
                 .mapToInt(w -> w.getWeight())
                 .sum();
```

See the class documentation for `Stream` and the package documentation for java.util.stream for additional specification of streams, stream operations, stream pipelines, and parallelism.

Since:

1.8

See Also:

`Stream`, java.util.stream

Nested Class Summary

Nested Classes

Modifier and Type	Interface	Description
static interface	**IntStream.Builder**	A mutable builder for an `IntStream`.
static interface	**IntStream.IntMapMultiConsume**	Represents an operation that accepts an `int`-valued argument and an IntConsumer, and returns no result.

Method Summary

All Methods | Static Methods | Instance Methods | Abstract Methods | Default Methods

Modifier and Type	Method	Description
boolean	allMatch (IntPredicate predicate)	Returns whether all elements of this stream match the provided predicate.
boolean	anyMatch (IntPredicate predicate)	Returns whether any elements of this stream match the provided predicate.
DoubleStream	asDoubleStream()	Returns a DoubleStream consisting of the elements of this stream, converted to double.
LongStream	asLongStream()	Returns a LongStream consisting of the elements of this stream, converted to long.
OptionalDouble	average()	Returns an OptionalDouble describing the arithmetic mean of elements of this stream, or an empty optional if this stream is empty.
Stream<Integer>	boxed()	Returns a Stream consisting of the elements of this stream, each boxed to an Integer.
static IntStream.Builder	builder()	Returns a builder for an IntStream.
<R> R	collect (Supplier<R> supplier, ObjIntConsumer<R> accumula BiConsumer<R,R> combiner)	Performs a mutable reduction operation on the elements of this stream.
static IntStream	concat(IntStream a, IntStream b)	Creates a lazily concatenated stream whose elements are all the elements of the first stream followed by all the elements of the second stream.
long	count()	Returns the count of elements in this stream.

IntStream	distinct()	Returns a stream consisting of the distinct elements of this stream.
default IntStream	dropWhile (IntPredicate predicate)	Returns, if this stream is ordered, a stream consisting of the remaining elements of this stream after dropping the longest prefix of elements that match the given predicate.
static IntStream	empty()	Returns an empty sequential IntStream.
IntStream	filter (IntPredicate predicate)	Returns a stream consisting of the elements of this stream that match the given predicate.
OptionalInt	findAny()	Returns an OptionalInt describing some element of the stream, or an empty OptionalInt if the stream is empty.
OptionalInt	findFirst()	Returns an OptionalInt describing the first element of this stream, or an empty OptionalInt if the stream is empty.
IntStream	flatMap(IntFunction<? extends IntStream> mapper)	Returns a stream consisting of the results of replacing each element of this stream with the contents of a mapped stream produced by applying the provided mapping function to each element.
void	forEach (IntConsumer action)	Performs an action for each element of this stream.
void	forEachOrdered (IntConsumer action)	Performs an action for each element of this stream, guaranteeing that each element is processed in encounter order for streams that have a defined encounter order.
static IntStream	generate(IntSupplier s)	Returns an infinite sequential unordered stream

		where each element is generated by the provided `IntSupplier`.
static **IntStream**	**iterate**(int seed, **IntPredicate** hasNext, **IntUnaryOperator** next)	Returns a sequential ordered `IntStream` produced by iterative application of the given `next` function to an initial element, conditioned on satisfying the given `hasNext` predicate.
static **IntStream**	**iterate**(int seed, **IntUnaryOperator** f)	Returns an infinite sequential ordered `IntStream` produced by iterative application of a function `f` to an initial element `seed`, producing a `Stream` consisting of `seed`, `f(seed)`, `f(f(seed))`, etc.
IntStream	**limit**(long maxSize)	Returns a stream consisting of the elements of this stream, truncated to be no longer than `maxSize` in length.
IntStream	**map** (**IntUnaryOperator** mapper)	Returns a stream consisting of the results of applying the given function to the elements of this stream.
default **IntStream**	**mapMulti** (**IntStream.IntMapMultiCons**	Returns a stream consisting of the results of replacing each element of this stream with multiple elements, specifically zero or more elements.
DoubleStream	**mapToDouble** (**IntToDoubleFunction** mappe	Returns a `DoubleStream` consisting of the results of applying the given function to the elements of this stream.
LongStream	**mapToLong** (**IntToLongFunction** mapper)	Returns a `LongStream` consisting of the results of applying the given function to the elements of this stream.
<U> **Stream**<U>	**mapToObj**(**IntFunction**<? extends U> mapper)	Returns an object-valued `Stream` consisting of the

		results of applying the given function to the elements of this stream.
OptionalInt	**max**()	Returns an OptionalInt describing the maximum element of this stream, or an empty optional if this stream is empty.
OptionalInt	**min**()	Returns an OptionalInt describing the minimum element of this stream, or an empty optional if this stream is empty.
boolean	**noneMatch** (**IntPredicate** predicate)	Returns whether no elements of this stream match the provided predicate.
static **IntStream**	**of**(int t)	Returns a sequential IntStream containing a single element.
static **IntStream**	**of**(int... values)	Returns a sequential ordered stream whose elements are the specified values.
IntStream	**peek**(**IntConsumer** action)	Returns a stream consisting of the elements of this stream, additionally performing the provided action on each element as elements are consumed from the resulting stream.
static **IntStream**	**range**(int startInclusive, int endExclusive)	Returns a sequential ordered IntStream from startInclusive (inclusive) to endExclusive (exclusive) by an incremental step of 1.
static **IntStream**	**rangeClosed** (int startInclusive, int endInclusive)	Returns a sequential ordered IntStream from startInclusive (inclusive) to endInclusive (inclusive) by an incremental step of 1.
int	**reduce**(int identity, **IntBinaryOperator** op)	Performs a reduction on the elements of this stream, using the provided identity value and an associative

		accumulation function, and returns the reduced value.
OptionalInt	**reduce** (**IntBinaryOperator** op)	Performs a reduction on the elements of this stream, using an associative accumulation function, and returns an `OptionalInt` describing the reduced value, if any.
IntStream	**skip**(long n)	Returns a stream consisting of the remaining elements of this stream after discarding the first `n` elements of the stream.
IntStream	**sorted**()	Returns a stream consisting of the elements of this stream in sorted order.
int	**sum**()	Returns the sum of elements in this stream.
IntSummaryStatistics	**summaryStatistics**()	Returns an `IntSummaryStatistics` describing various summary data about the elements of this stream.
default **IntStream**	**takeWhile** (**IntPredicate** predicate)	Returns, if this stream is ordered, a stream consisting of the longest prefix of elements taken from this stream that match the given predicate.
int[]	**toArray**()	Returns an array containing the elements of this stream.

Methods declared in interface java.util.stream.BaseStream

close, isParallel, iterator, onClose, parallel, sequential, spliterator, unordered

Method Details

filter

IntStream filter(IntPredicate predicate)

Returns a stream consisting of the elements of this stream that match the given predicate.

This is an intermediate operation.

Parameters:

`predicate` - a non-interfering, stateless predicate to apply to each element to determine if it should be included

Returns:

the new stream

map

```
IntStream map(IntUnaryOperator mapper)
```

Returns a stream consisting of the results of applying the given function to the elements of this stream.

This is an intermediate operation.

Parameters:

`mapper` - a non-interfering, stateless function to apply to each element

Returns:

the new stream

mapToObj

```
<U> Stream<U> mapToObj(IntFunction<? extends U> mapper)
```

Returns an object-valued `Stream` consisting of the results of applying the given function to the elements of this stream.

This is an intermediate operation.

Type Parameters:

`U` - the element type of the new stream

Parameters:

`mapper` - a non-interfering, stateless function to apply to each element

Returns:

the new stream

mapToLong

```
LongStream mapToLong(IntToLongFunction mapper)
```

Returns a `LongStream` consisting of the results of applying the given function to the elements of this stream.

This is an intermediate operation.

Parameters:

`mapper` - a non-interfering, stateless function to apply to each element

Returns:

the new stream

mapToDouble

```
DoubleStream mapToDouble(IntToDoubleFunction mapper)
```

Returns a `DoubleStream` consisting of the results of applying the given function to the elements of this stream.

This is an intermediate operation.

Parameters:

`mapper` - a non-interfering, stateless function to apply to each element

Returns:

the new stream

flatMap

```
IntStream flatMap(IntFunction<? extends IntStream> mapper)
```

Returns a stream consisting of the results of replacing each element of this stream with the contents of a mapped stream produced by applying the provided mapping function to each element. Each mapped stream is `closed` after its contents have been placed into this stream. (If a mapped stream is `null` an empty stream is used, instead.)

This is an intermediate operation.

Parameters:

`mapper` - a non-interfering, stateless function to apply to each element which produces an `IntStream` of new values

Returns:

the new stream

See Also:

`Stream.flatMap(Function)`

mapMulti

```
default IntStream mapMulti(IntStream.IntMapMultiConsumer mapper)
```

Returns a stream consisting of the results of replacing each element of this stream with multiple elements, specifically zero or more elements. Replacement is performed by applying the provided mapping function to each element in conjunction with a consumer argument that accepts replacement elements. The mapping function calls the consumer zero or more times to provide the replacement elements.

This is an intermediate operation.

If the consumer argument is used outside the scope of its application to the mapping function, the results are undefined.

Implementation Requirements:

The default implementation invokes `flatMap` on this stream, passing a function that behaves as follows. First, it calls the mapper function with an `IntConsumer` that accumulates replacement elements into a newly created internal buffer. When the mapper function returns, it creates an `IntStream` from the internal buffer. Finally, it returns this stream to `flatMap`.

Parameters:

`mapper` - a non-interfering, stateless function that generates replacement elements

Returns:

the new stream

Since:

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See Also:

`Stream.mapMulti`

distinct

```
IntStream distinct()
```

Returns a stream consisting of the distinct elements of this stream.

This is a stateful intermediate operation.

Returns:

the new stream

sorted

```
IntStream sorted()
```

Returns a stream consisting of the elements of this stream in sorted order.

This is a stateful intermediate operation.

Returns:

the new stream

peek

```
IntStream peek(IntConsumer action)
```

Returns a stream consisting of the elements of this stream, additionally performing the provided action on each element as elements are consumed from the resulting stream.

This is an intermediate operation.

For parallel stream pipelines, the action may be called at whatever time and in whatever thread the element is made available by the upstream operation. If the action modifies shared state, it is responsible for providing the required synchronization.

API Note:

This method exists mainly to support debugging, where you want to see the elements as they flow past a certain point in a pipeline:

```
IntStream.of(1, 2, 3, 4)
    .filter(e -> e > 2)
    .peek(e -> System.out.println("Filtered value: " + e))
    .map(e -> e * e)
    .peek(e -> System.out.println("Mapped value: " + e))
    .sum();
```

In cases where the stream implementation is able to optimize away the production of some or all the elements (such as with short-circuiting operations like `findFirst`, or in the example described in `count()`), the action will not be invoked for those elements.

Parameters:

`action` - a non-interfering action to perform on the elements as they are consumed from the stream

Returns:

the new stream

limit

`IntStream limit(long maxSize)`

Returns a stream consisting of the elements of this stream, truncated to be no longer than `maxSize` in length.

This is a short-circuiting stateful intermediate operation.

API Note:

While `limit()` is generally a cheap operation on sequential stream pipelines, it can be quite expensive on ordered parallel pipelines, especially for large values of `maxSize`, since `limit(n)` is constrained to return not just any *n* elements, but the *first n* elements in the encounter order. Using an unordered stream source (such as `generate(IntSupplier)`) or removing the ordering constraint with `BaseStream.unordered()` may result in significant speedups of `limit()` in parallel pipelines, if the semantics of your situation permit. If consistency with encounter order is required, and you are experiencing poor performance or memory utilization with `limit()` in parallel pipelines, switching to sequential execution with `BaseStream.sequential()` may improve performance.

Parameters:

`maxSize` - the number of elements the stream should be limited to

Returns:

the new stream

Throws:

`IllegalArgumentException` - if `maxSize` is negative

skip

```
IntStream skip(long n)
```

Returns a stream consisting of the remaining elements of this stream after discarding the first `n` elements of the stream. If this stream contains fewer than `n` elements then an empty stream will be returned.

This is a stateful intermediate operation.

API Note:

While `skip()` is generally a cheap operation on sequential stream pipelines, it can be quite expensive on ordered parallel pipelines, especially for large values of `n`, since `skip(n)` is constrained to skip not just any *n* elements, but the *first n* elements in the encounter order. Using an unordered stream source (such as `generate(IntSupplier)`) or removing the ordering constraint with `BaseStream.unordered()` may result in significant speedups of `skip()` in parallel pipelines, if the semantics of your situation permit. If consistency with encounter order is required, and you are experiencing poor performance or memory utilization with `skip()` in parallel pipelines, switching to sequential execution with `BaseStream.sequential()` may improve performance.

Parameters:

`n` - the number of leading elements to skip

Returns:

the new stream

Throws:

`IllegalArgumentException` - if `n` is negative

takeWhile

```
default IntStream takeWhile(IntPredicate predicate)
```

Returns, if this stream is ordered, a stream consisting of the longest prefix of elements taken from this stream that match the given predicate. Otherwise returns, if this stream is unordered, a stream consisting of a subset of elements taken from this stream that match the given predicate.

If this stream is ordered then the longest prefix is a contiguous sequence of elements of this stream that match the given predicate. The first element of the sequence is the first element of this stream, and the element immediately following the last element of the sequence does not match the given predicate.

If this stream is unordered, and some (but not all) elements of this stream match the given predicate, then the behavior of this operation is nondeterministic; it is free to take any subset of matching elements (which includes the empty set).

Independent of whether this stream is ordered or unordered if all elements of this stream match the given predicate then this operation takes all elements (the result is

the same as the input), or if no elements of the stream match the given predicate then no elements are taken (the result is an empty stream).

This is a short-circuiting stateful intermediate operation.

API Note:

While `takeWhile()` is generally a cheap operation on sequential stream pipelines, it can be quite expensive on ordered parallel pipelines, since the operation is constrained to return not just any valid prefix, but the longest prefix of elements in the encounter order. Using an unordered stream source (such as `generate(IntSupplier)`) or removing the ordering constraint with `BaseStream.unordered()` may result in significant speedups of `takeWhile()` in parallel pipelines, if the semantics of your situation permit. If consistency with encounter order is required, and you are experiencing poor performance or memory utilization with `takeWhile()` in parallel pipelines, switching to sequential execution with `BaseStream.sequential()` may improve performance.

Implementation Requirements:

The default implementation obtains the `spliterator` of this stream, wraps that spliterator so as to support the semantics of this operation on traversal, and returns a new stream associated with the wrapped spliterator. The returned stream preserves the execution characteristics of this stream (namely parallel or sequential execution as per `BaseStream.isParallel()`) but the wrapped spliterator may choose to not support splitting. When the returned stream is closed, the close handlers for both the returned and this stream are invoked.

Parameters:

`predicate` - a non-interfering, stateless predicate to apply to elements to determine the longest prefix of elements.

Returns:

the new stream

Since:

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dropWhile

```
default IntStream dropWhile(IntPredicate predicate)
```

Returns, if this stream is ordered, a stream consisting of the remaining elements of this stream after dropping the longest prefix of elements that match the given predicate. Otherwise returns, if this stream is unordered, a stream consisting of the remaining elements of this stream after dropping a subset of elements that match the given predicate.

If this stream is ordered then the longest prefix is a contiguous sequence of elements of this stream that match the given predicate. The first element of the sequence is the first element of this stream, and the element immediately following the last element of the sequence does not match the given predicate.

If this stream is unordered, and some (but not all) elements of this stream match the given predicate, then the behavior of this operation is nondeterministic; it is free to drop any subset of matching elements (which includes the empty set).

Independent of whether this stream is ordered or unordered if all elements of this stream match the given predicate then this operation drops all elements (the result is an empty stream), or if no elements of the stream match the given predicate then no elements are dropped (the result is the same as the input).

This is a stateful intermediate operation.

API Note:

While `dropWhile()` is generally a cheap operation on sequential stream pipelines, it can be quite expensive on ordered parallel pipelines, since the operation is constrained to return not just any valid prefix, but the longest prefix of elements in the encounter order. Using an unordered stream source (such as `generate(IntSupplier)`) or removing the ordering constraint with `BaseStream.unordered()` may result in significant speedups of `dropWhile()` in parallel pipelines, if the semantics of your situation permit. If consistency with encounter order is required, and you are experiencing poor performance or memory utilization with `dropWhile()` in parallel pipelines, switching to sequential execution with `BaseStream.sequential()` may improve performance.

Implementation Requirements:

The default implementation obtains the `spliterator` of this stream, wraps that spliterator so as to support the semantics of this operation on traversal, and returns a new stream associated with the wrapped spliterator. The returned stream preserves the execution characteristics of this stream (namely parallel or sequential execution as per `BaseStream.isParallel()`) but the wrapped spliterator may choose to not support splitting. When the returned stream is closed, the close handlers for both the returned and this stream are invoked.

Parameters:

`predicate` - a non-interfering, stateless predicate to apply to elements to determine the longest prefix of elements.

Returns:

the new stream

Since:

9

forEach

```
void forEach(IntConsumer action)
```

Performs an action for each element of this stream.

This is a terminal operation.

For parallel stream pipelines, this operation does *not* guarantee to respect the encounter order of the stream, as doing so would sacrifice the benefit of parallelism. For any given element, the action may be performed at whatever time and in whatever thread the library chooses. If the action accesses shared state, it is responsible for providing the required synchronization.

Parameters:

`action` - a non-interfering action to perform on the elements

forEachOrdered

```
void forEachOrdered(IntConsumer action)
```

Performs an action for each element of this stream, guaranteeing that each element is processed in encounter order for streams that have a defined encounter order.

This is a terminal operation.

Parameters:

`action` - a non-interfering action to perform on the elements

See Also:

`forEach(IntConsumer)`

toArray

```
int[] toArray()
```

Returns an array containing the elements of this stream.

This is a terminal operation.

Returns:

an array containing the elements of this stream

reduce

```
int reduce(int identity,
           IntBinaryOperator op)
```

Performs a reduction on the elements of this stream, using the provided identity value and an associative accumulation function, and returns the reduced value. This is equivalent to:

```
int result = identity;
for (int element : this stream)
    result = accumulator.applyAsInt(result, element)
return result;
```

but is not constrained to execute sequentially.

The `identity` value must be an identity for the accumulator function. This means that for all `x`, `accumulator.apply(identity, x)` is equal to `x`. The `accumulator` function must be an associative function.

This is a terminal operation.

API Note:

Sum, min and max are all special cases of reduction that can be expressed using this method. For example, summing a stream can be expressed as:

```
int sum = integers.reduce(0, (a, b) -> a+b);
```

or more compactly:

```
int sum = integers.reduce(0, Integer::sum);
```

While this may seem a more roundabout way to perform an aggregation compared to simply mutating a running total in a loop, reduction operations parallelize more gracefully, without needing additional synchronization and with greatly reduced risk of data races.

Parameters:

`identity` - the identity value for the accumulating function

`op` - an associative, non-interfering, stateless function for combining two values

Returns:

the result of the reduction

See Also:

`sum()`, `min()`, `max()`, `average()`

reduce

`OptionalInt reduce(IntBinaryOperator op)`

Performs a reduction on the elements of this stream, using an associative accumulation function, and returns an `OptionalInt` describing the reduced value, if any. This is equivalent to:

```
boolean foundAny = false;
int result = null;
for (int element : this stream) {
    if (!foundAny) {
        foundAny = true;
        result = element;
    }
    else
        result = accumulator.applyAsInt(result, element);
}
return foundAny ? OptionalInt.of(result) : OptionalInt.empty();
```

but is not constrained to execute sequentially.

The `accumulator` function must be an associative function.

This is a terminal operation.

Parameters:

`op` - an associative, non-interfering, stateless function for combining two values

Returns:

the result of the reduction

See Also:

`reduce(int, IntBinaryOperator)`

collect

```
<R> R collect(Supplier<R> supplier,
              ObjIntConsumer<R> accumulator,
              BiConsumer<R,R> combiner)
```

Performs a mutable reduction operation on the elements of this stream. A mutable reduction is one in which the reduced value is a mutable result container, such as an `ArrayList`, and elements are incorporated by updating the state of the result rather than by replacing the result. This produces a result equivalent to:

```
R result = supplier.get();
for (int element : this stream)
    accumulator.accept(result, element);
return result;
```

Like `reduce(int, IntBinaryOperator)`, `collect` operations can be parallelized without requiring additional synchronization.

This is a terminal operation.

Type Parameters:

R - the type of the mutable result container

Parameters:

`supplier` - a function that creates a new mutable result container. For a parallel execution, this function may be called multiple times and must return a fresh value each time.

`accumulator` - an associative, non-interfering, stateless function that must fold an element into a result container.

`combiner` - an associative, non-interfering, stateless function that accepts two partial result containers and merges them, which must be compatible with the accumulator function. The combiner function must fold the elements from the second result container into the first result container.

Returns:

the result of the reduction

See Also:

`Stream.collect(Supplier, BiConsumer, BiConsumer)`

sum

```
int sum()
```

Returns the sum of elements in this stream. This is a special case of a reduction and is equivalent to:

```
return reduce(0, Integer::sum);
```

This is a terminal operation.

Returns:

the sum of elements in this stream

min

`OptionalInt min()`

Returns an `OptionalInt` describing the minimum element of this stream, or an empty optional if this stream is empty. This is a special case of a reduction and is equivalent to:

```
return reduce(Integer::min);
```

This is a terminal operation.

Returns:

an `OptionalInt` containing the minimum element of this stream, or an empty `OptionalInt` if the stream is empty

max

`OptionalInt max()`

Returns an `OptionalInt` describing the maximum element of this stream, or an empty optional if this stream is empty. This is a special case of a reduction and is equivalent to:

```
return reduce(Integer::max);
```

This is a terminal operation.

Returns:

an `OptionalInt` containing the maximum element of this stream, or an empty `OptionalInt` if the stream is empty

count

`long count()`

Returns the count of elements in this stream. This is a special case of a reduction and is equivalent to:

```
return mapToLong(e -> 1L).sum();
```

This is a terminal operation.

API Note:

An implementation may choose to not execute the stream pipeline (either sequentially or in parallel) if it is capable of computing the count directly from the stream source. In such cases no source elements will be traversed and no intermediate operations will be evaluated. Behavioral parameters with side-effects, which are strongly discouraged except for harmless cases such as debugging, may be affected. For example, consider the following stream:

```
IntStream s = IntStream.of(1, 2, 3, 4);
long count = s.peek(System.out::println).count();
```

The number of elements covered by the stream source is known and the intermediate operation, `peek`, does not inject into or remove elements from the stream (as may be the case for `flatMap` or `filter` operations). Thus the count is 4 and there is no need to execute the pipeline and, as a side-effect, print out the elements.

Returns:

the count of elements in this stream

average

`OptionalDouble average()`

Returns an `OptionalDouble` describing the arithmetic mean of elements of this stream, or an empty optional if this stream is empty. This is a special case of a reduction.

This is a terminal operation.

Returns:

an `OptionalDouble` containing the average element of this stream, or an empty optional if the stream is empty

summaryStatistics

`IntSummaryStatistics summaryStatistics()`

Returns an `IntSummaryStatistics` describing various summary data about the elements of this stream. This is a special case of a reduction.

This is a terminal operation.

Returns:

an `IntSummaryStatistics` describing various summary data about the elements of this stream

anyMatch

```
boolean anyMatch(IntPredicate predicate)
```

Returns whether any elements of this stream match the provided predicate. May not evaluate the predicate on all elements if not necessary for determining the result. If the stream is empty then `false` is returned and the predicate is not evaluated.

This is a short-circuiting terminal operation.

API Note:

This method evaluates the *existential quantification* of the predicate over the elements of the stream (for some x P(x)).

Parameters:

`predicate` - a non-interfering, stateless predicate to apply to elements of this stream

Returns:

`true` if any elements of the stream match the provided predicate, otherwise `false`

allMatch

```
boolean allMatch(IntPredicate predicate)
```

Returns whether all elements of this stream match the provided predicate. May not evaluate the predicate on all elements if not necessary for determining the result. If the stream is empty then `true` is returned and the predicate is not evaluated.

This is a short-circuiting terminal operation.

API Note:

This method evaluates the *universal quantification* of the predicate over the elements of the stream (for all x P(x)). If the stream is empty, the quantification is said to be *vacuously satisfied* and is always `true` (regardless of P(x)).

Parameters:

`predicate` - a non-interfering, stateless predicate to apply to elements of this stream

Returns:

`true` if either all elements of the stream match the provided predicate or the stream is empty, otherwise `false`

noneMatch

```
boolean noneMatch(IntPredicate predicate)
```

Returns whether no elements of this stream match the provided predicate. May not evaluate the predicate on all elements if not necessary for determining the result. If the stream is empty then `true` is returned and the predicate is not evaluated.

This is a short-circuiting terminal operation.

API Note:

This method evaluates the *universal quantification* of the negated predicate over the elements of the stream (for all x ~P(x)). If the stream is empty, the quantification is said to be vacuously satisfied and is always `true`, regardless of P(x).

Parameters:

`predicate` - a non-interfering, stateless predicate to apply to elements of this stream

Returns:

`true` if either no elements of the stream match the provided predicate or the stream is empty, otherwise `false`

findFirst

```
OptionalInt findFirst()
```

Returns an `OptionalInt` describing the first element of this stream, or an empty `OptionalInt` if the stream is empty. If the stream has no encounter order, then any element may be returned.

This is a short-circuiting terminal operation.

Returns:

an `OptionalInt` describing the first element of this stream, or an empty `OptionalInt` if the stream is empty

findAny

```
OptionalInt findAny()
```

Returns an `OptionalInt` describing some element of the stream, or an empty `OptionalInt` if the stream is empty.

This is a short-circuiting terminal operation.

The behavior of this operation is explicitly nondeterministic; it is free to select any element in the stream. This is to allow for maximal performance in parallel operations; the cost is that multiple invocations on the same source may not return the same result. (If a stable result is desired, use `findFirst()` instead.)

Returns:

an `OptionalInt` describing some element of this stream, or an empty `OptionalInt` if the stream is empty

See Also:

`findFirst()`

asLongStream

```
LongStream asLongStream()
```

Returns a `LongStream` consisting of the elements of this stream, converted to `long`.

This is an intermediate operation.

Returns:

a `LongStream` consisting of the elements of this stream, converted to `long`

asDoubleStream

`DoubleStream asDoubleStream()`

Returns a `DoubleStream` consisting of the elements of this stream, converted to `double`.

This is an intermediate operation.

Returns:

a `DoubleStream` consisting of the elements of this stream, converted to `double`

boxed

`Stream<Integer> boxed()`

Returns a `Stream` consisting of the elements of this stream, each boxed to an `Integer`.

This is an intermediate operation.

Returns:

a `Stream` consistent of the elements of this stream, each boxed to an `Integer`

builder

`static IntStream.Builder builder()`

Returns a builder for an `IntStream`.

Returns:

a stream builder

empty

`static IntStream empty()`

Returns an empty sequential `IntStream`.

Returns:

an empty sequential stream

of

`static IntStream of(int t)`

Returns a sequential `IntStream` containing a single element.

Parameters:

`t` - the single element

Returns:

a singleton sequential stream

of

```
static IntStream of(int... values)
```

Returns a sequential ordered stream whose elements are the specified values.

Parameters:

`values` - the elements of the new stream

Returns:

the new stream

iterate

```
static IntStream iterate(int seed,
                         IntUnaryOperator f)
```

Returns an infinite sequential ordered `IntStream` produced by iterative application of a function `f` to an initial element `seed`, producing a `Stream` consisting of `seed`, `f(seed)`, `f(f(seed))`, etc.

The first element (position `0`) in the `IntStream` will be the provided `seed`. For `n > 0`, the element at position `n`, will be the result of applying the function `f` to the element at position `n - 1`.

The action of applying `f` for one element *happens-before* the action of applying `f` for subsequent elements. For any given element the action may be performed in whatever thread the library chooses.

Parameters:

`seed` - the initial element

`f` - a function to be applied to the previous element to produce a new element

Returns:

a new sequential `IntStream`

iterate

```
static IntStream iterate(int seed,
                         IntPredicate hasNext,
                         IntUnaryOperator next)
```

Returns a sequential ordered `IntStream` produced by iterative application of the given `next` function to an initial element, conditioned on satisfying the given `hasNext`

predicate. The stream terminates as soon as the `hasNext` predicate returns false.

`IntStream.iterate` should produce the same sequence of elements as produced by the corresponding for-loop:

```
for (int index=seed; hasNext.test(index); index = next.applyAsInt(index)) {
    ...
}
```

The resulting sequence may be empty if the `hasNext` predicate does not hold on the seed value. Otherwise the first element will be the supplied `seed` value, the next element (if present) will be the result of applying the `next` function to the `seed` value, and so on iteratively until the `hasNext` predicate indicates that the stream should terminate.

The action of applying the `hasNext` predicate to an element *happens-before* the action of applying the `next` function to that element. The action of applying the `next` function for one element *happens-before* the action of applying the `hasNext` predicate for subsequent elements. For any given element an action may be performed in whatever thread the library chooses.

Parameters:

`seed` - the initial element

`hasNext` - a predicate to apply to elements to determine when the stream must terminate.

`next` - a function to be applied to the previous element to produce a new element

Returns:

a new sequential `IntStream`

Since:

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generate

```
static IntStream generate(IntSupplier s)
```

Returns an infinite sequential unordered stream where each element is generated by the provided `IntSupplier`. This is suitable for generating constant streams, streams of random elements, etc.

Parameters:

`s` - the `IntSupplier` for generated elements

Returns:

a new infinite sequential unordered `IntStream`

range

```
static IntStream range(int startInclusive,
                       int endExclusive)
```

Returns a sequential ordered `IntStream` from `startInclusive` (inclusive) to `endExclusive` (exclusive) by an incremental step of `1`.

API Note:

An equivalent sequence of increasing values can be produced sequentially using a `for` loop as follows:

```
    for (int i = startInclusive; i < endExclusive ; i++) { ... }
```

Parameters:

`startInclusive` - the (inclusive) initial value

`endExclusive` - the exclusive upper bound

Returns:

a sequential `IntStream` for the range of `int` elements

rangeClosed

```
static IntStream rangeClosed(int startInclusive,
                             int endInclusive)
```

Returns a sequential ordered `IntStream` from `startInclusive` (inclusive) to `endInclusive` (inclusive) by an incremental step of `1`.

API Note:

An equivalent sequence of increasing values can be produced sequentially using a `for` loop as follows:

```
    for (int i = startInclusive; i <= endInclusive ; i++) { ... }
```

Parameters:

`startInclusive` - the (inclusive) initial value

`endInclusive` - the inclusive upper bound

Returns:

a sequential `IntStream` for the range of `int` elements

concat

```
static IntStream concat(IntStream a,
                        IntStream b)
```

Creates a lazily concatenated stream whose elements are all the elements of the first stream followed by all the elements of the second stream. The resulting stream is ordered if both of the input streams are ordered, and parallel if either of the input streams is parallel. When the resulting stream is closed, the close handlers for both input streams are invoked.

This method operates on the two input streams and binds each stream to its source. As a result subsequent modifications to an input stream source may not be reflected in the concatenated stream result.

API Note:

To preserve optimization opportunities this method binds each stream to its source and accepts only two streams as parameters. For example, the exact size of the concatenated stream source can be computed if the exact size of each input stream source is known. To concatenate more streams without binding, or without nested calls to this method, try creating a stream of streams and flat-mapping with the identity function, for example:

```
IntStream concat = Stream.of(s1, s2, s3, s4).flatMapToInt(s -> s);
```

Implementation Note:

Use caution when constructing streams from repeated concatenation. Accessing an element of a deeply concatenated stream can result in deep call chains, or even `StackOverflowError`.

Parameters:

a - the first stream

b - the second stream

Returns:

the concatenation of the two input streams
